

Text-Driven Automated Multi-Character Voice Generation for Narrative Content

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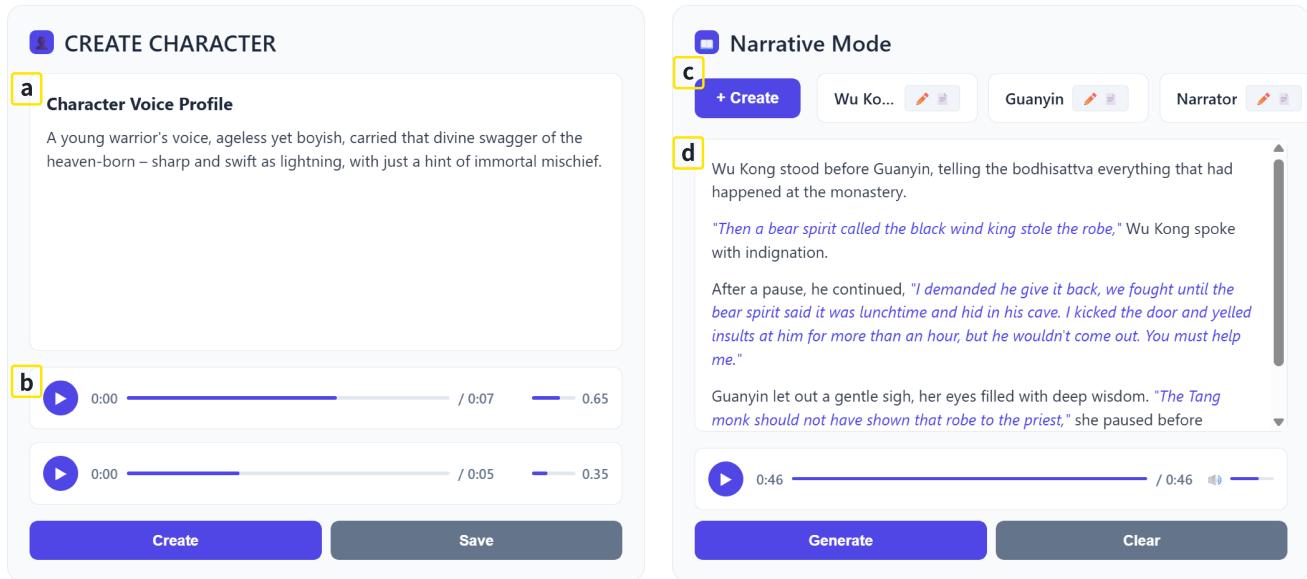


Figure 1: User interface overview: (a) Character creation area with text input for voice description; (b) Audio mixing panel for fine-tuning voice characteristics; (c) Character list displaying created voices; (d) Script input area with automated dialogue detection.

Abstract

We present a text-driven system that automatically generates distinctive voices for multiple characters in narrative content through natural language descriptions. Our interactive platform converts textual character descriptions into unique voice configurations

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and intelligently assigns voices to different characters in scripts without manual intervention. The system features an intuitive interface that combines text-driven voice generation with real-time audio mixing capabilities, enabling content creators to produce professional-quality multi-character audio narratives efficiently. Users can describe desired voice characteristics in natural language (e.g., "an elderly wizard with a deep, mysterious voice") and the system automatically identifies dialogue in scripts to assign corresponding voices. This demo showcases the complete automated workflow from character description to final multi-character narrative audio production, demonstrating significant improvements in efficiency over traditional voice production methods.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools.**

Keywords

Text-to-Speech, Character Voice Generation, Voice Customization, Narrative Content Creation

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1 INTRODUCTION

The digital narrative content industry has experienced unprecedented growth, with the global audiobook market reaching USD 6.83 billion in 2023[6] and demonstrating consistent 20% year-over-year growth[12]. Voice quality significantly impacts audience engagement[5, 10], yet current voice production approaches face substantial limitations. Professional voice acting delivers high-quality results but involves complex production processes and high costs that limit scalability, while single AI voice narration offers convenience but compromises narrative expression through monotonous delivery. Existing multi-voice AI systems like iFLYTEK Smart Creation¹ require extensive manual voice selection and script annotation, creating bottlenecks for multi-character content creation. Although recent advances in Text-to-Speech (TTS) technology and Large Language Models (LLMs) have demonstrated remarkable capabilities[3, 4, 14], significant gaps remain between technical possibilities and creative needs. Voice cloning technologies excel at replicating existing voices but cannot create new voices that align with creators' artistic visions, and effectively transforming natural language character descriptions into usable voice parameters remains challenging[11].

We present VoiceForge, an interactive system that automatically converts textual character descriptions into distinctive voice configurations and enables automated multi-character narrative production. Our system addresses three key challenges: (1) mapping natural language descriptions to stable voice characteristics, (2) automating voice assignment in multi-character scripts, and (3) maintaining voice consistency across extended narratives.

Our main contributions include:

- (1) A novel text-to-voice mapping framework that automatically converts character descriptions into stable voice characteristics using an 8-dimensional acoustic feature space;
- (2) An intelligent script analysis and voice assignment system that automatically identifies dialogue and assigns appropriate character voices;
- (3) A hybrid interactive interface supporting both text-driven generation and real-time audio mixing for flexible voice customization;
- (4) A complete demonstration of automated multi-character narrative audio production from character description to final audio output.

¹<https://peiyin.xunfei.cn/>

2 RELATED WORK

2.1 Voice Customization and Multi-Character Systems

Voice customization has traditionally been limited by high costs and implementation complexities. Commercial applications like Voicemod² offer basic real-time voice transformation through pre-set filters, but provide limited options often unsuitable for narrative content[2, 8]. More sophisticated approaches like AVOCUS[1] combine voice search with acoustic parameter adjustment, while specialized systems have explored game character voice recommendations using acoustic feature estimation[9]. However, these solutions struggle to achieve rapid, flexible voice generation that effectively utilizes textual information for multi-character customization.

2.2 Text-to-Speech and Voice Synthesis

The evolution of TTS technology has progressed from the early sequence-to-sequence models[13] to attention-based architectures such as Tacotron[15], significantly improving synthesized speech naturalness. Recent work include HuBERT's self-supervised speech representation learning, enhancing voice quality consistency across extended content[7].

Voice cloning technology has demonstrated remarkable capabilities, with systems like VALL-E and YourTTS[3, 14] achieving zero-shot voice synthesis from brief audio samples. However, these systems face challenges in maintaining stability across longer narratives and handling emotional variations. Recent research has explored voice characteristic control through text descriptions, exemplified by CosyVoice[4]. While CosyVoice-300M-Instruct generates voice characteristics from textual input, it exhibits timbre shifts during extended synthesis and lacks fine-grained control. In contrast, CosyVoice-300M demonstrates superior stability and naturalness, making it an ideal foundation for consistent multi-character voice generation, though it requires external text-to-voice mapping mechanisms.

3 INTERACTION

VoiceForge provides an intuitive workflow designed for efficient multi-character voice creation and narrative audio production. The character creation process begins in the text input area (Figure 1(a)), where users describe desired voice characteristics using natural language such as "a young, energetic female voice with a slight rasp" or "an elderly wizard with a deep, mysterious voice." The system immediately generates a speech sample with test text, providing real-time feedback that allows users to iteratively refine their descriptions until achieving the desired voice.

For cases where text description alone cannot achieve the intended voice characteristics, the system offers an auxiliary audio mixing control panel (Figure 1(b)) that allows users to blend 1-3 reference audio samples with adjustable weights for fine-tuned customization. This hybrid approach bridges the gap between natural language expression and precise voice control.

Created characters appear in the character list area (Figure 1(c)), where users can rename or delete them. The system enforces exact name matching between characters in this list and character

²<https://www.voicemod.net/>

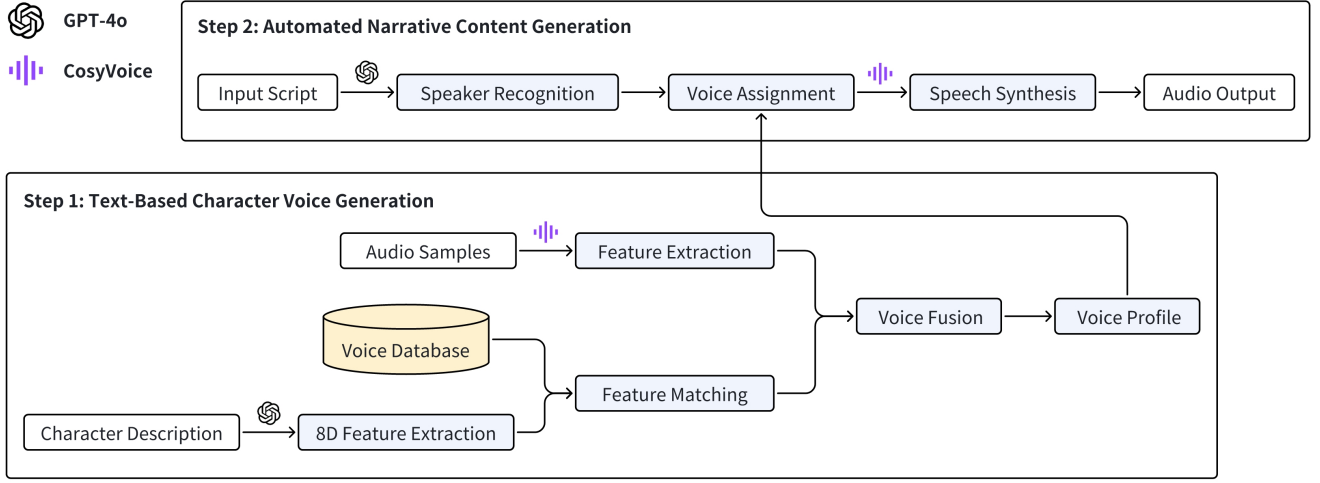


Figure 2: System architecture diagram showing the two-stage process of VoiceForge.

names in scripts to ensure proper voice assignment. After character creation, users input narrative text in the script input area (Figure 1(d)). The system automatically identifies dialogue content and matches it with corresponding character voices to generate the final multi-character audio output.

4 IMPLEMENTATION

As illustrated in Figure 2, VoiceForge achieves automated generation from character text descriptions to complete narrative audio content through a two-stage architecture. The first stage focuses on accurately converting character descriptions into stable voice characteristics, while the second stage implements automated production of multi-character narrative content.

The core innovation lies in converting natural language descriptions into stable voice characteristics through an 8-dimensional acoustic feature space. We leverage GPT-4o³'s natural language understanding to parse character descriptions into normalized feature vectors $F = \{f_1, \dots, f_8\}$, including gender, age, pitch, warmth, clarity, power, thickness, and smoothness. Each dimension is normalized to $[0, 1]$ to ensure consistent feature scaling across different voice characteristics.

Our system maintains a curated database of 40 professional voice samples, each associated with both a 192-dimensional speaker embedding vector from CosyVoice-300M's encoder and a manually annotated 8-dimensional feature vector. Feature matching employs weighted similarity calculation using LASSO regression-learned importance weights $W = \{w_1, \dots, w_8\}$:

$$\text{sim}(F, D_i) = \sum_{j=1}^8 w_j (f_j - d_{ij})^2 \quad (1)$$

where w_j represents the importance weight of the j -th feature dimension, ensuring perceptually significant features have greater

influence on matching. The system identifies the 3 most similar voice profiles for subsequent fusion.

The final voice profile is generated through weighted combination of either the 3 most similar database voices or user-provided audio samples. For text-driven input, weighting coefficients are determined based on vector distances in the acoustic feature space, with closer matches having stronger influence on final voice characteristics. For users choosing direct audio input, the system processes up to 3 reference audio samples through CosyVoice-300M's voice encoder, converting each into speaker embedding vectors that maintain compatibility with our fusion framework.

The second stage implements automated narrative content generation through intelligent script analysis using GPT-4o's language understanding capabilities. The system performs deep parsing of input scripts, accurately identifying speakers and their corresponding dialogue content across various script formats and dialogue patterns. Based on recognition results, the system passes each speaker's voice embedding vector along with their dialogue text to CosyVoice-300M for speech synthesis, generating voice segments with personalized characteristics. Finally, all audio segments are concatenated according to the script's chronological order to create complete narrative audio content, significantly reducing manual effort while maintaining voice consistency throughout the narrative.

5 CONCLUSION

This paper presents VoiceForge, a novel text-driven character voice generation system that bridges the gap between textual character descriptions and voice synthesis. Through its innovative approach combining natural language processing, acoustic feature mapping, and automated narrative content generation, VoiceForge demonstrates significant advantages in both voice quality and production efficiency. The system's intuitive interface and automated workflow significantly reduce the barriers to creating high-quality voiced narrative content. While certain limitations remain, particularly in emotional expression and dialect support, VoiceForge represents

³<https://openai.com/index/hello-gpt-4o/>

a significant step forward in making professional-quality voice content creation more accessible to a broader range of creators.

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